Week 3 Example 4

Fernaldy

4 October 2026

ANOVA

Scenario: A student trying to get some practice in carrying out ANOVA in SPSS decided to compare weights (kg) between three independent groups (A, B & C) with a null hypothesis (H0) that there is no difference in mean weight between groups in the population. The student first plotted histograms of weight for the 3 groups

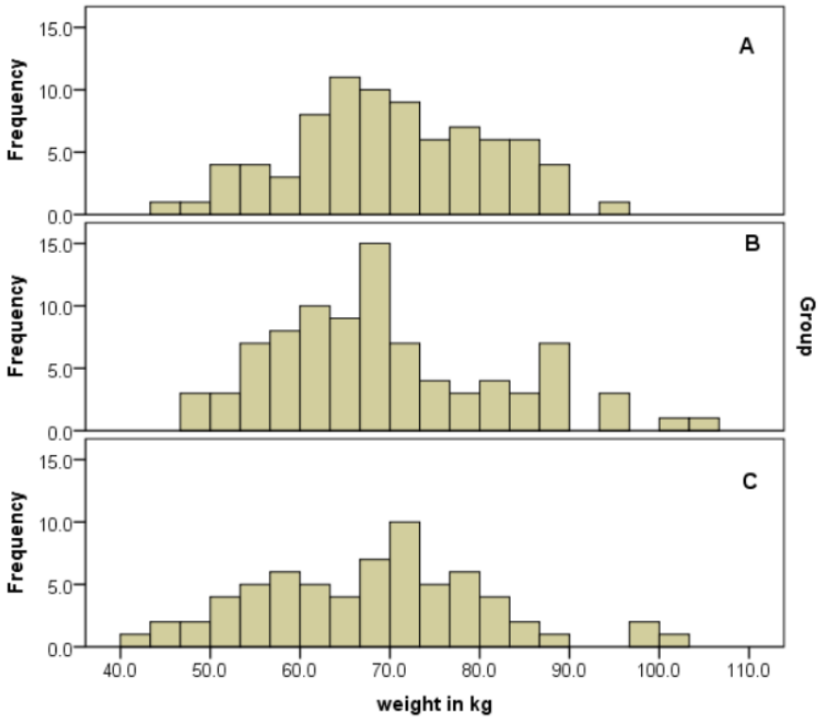


Figure 1: Histograms of weight for the 3 groups

Q1 - Comment on the distribution of weight in the 3 groups. Which assumption is being checked here?

The assumption being checked is for the normality in the distribution.
The distribution of weights in the three groups are roughly normal.

Q2a - What was the sample size? Comment on the size of the sample with respect to carrying out ANOVA.

Descriptives

weight in kg

	N	Mean	Std. Deviation	Std. Error	95% Confidence Interval for Mean		Minimum	Maximum
					Lower Bound	Upper Bound		
A	81	70.1341	11.03751	1.22639	67.6935	72.5747	44.91	93.44
B	88	69.2822	13.03126	1.38914	66.5211	72.0432	46.72	105.24
C	67	68.0064	12.87219	1.57259	64.8666	71.1462	42.18	101.61
Total	236	69.2124	12.31197	.80144	67.6334	70.7913	42.18	105.24

Percentiles

			Percentiles		
Group			25	50	75
Weighted Average(Definition 1)	weight in kg	A	62.8250	69.8500	78.2450
		B	60.3300	66.6800	78.5850
		C	58.0600	69.8500	75.3000
Tukey's Hinges	weight in kg	A	63.0500	69.8500	78.0200
		B	60.3300	66.6800	77.7900
		C	58.2850	69.8500	75.0700

Figure 2: Descriptive stats of weight for the 3 groups

The total sample size (all three groups) is 236, with

- 81 samples from group A
- 88 samples from group B
- 67 samples from group C

Furthermore, the sample size is also roughly balanced and each group has more than enough sample.

Q2b - Which descriptives should be presented? Write them out

The mean, sample size, and standard deviation should be presented for each group and the overall.

	N	Mean	SD
A	81	70.13	11.04
B	88	69.28	13.03
C	67	68.01	12.87
Total	236	69.21	12.31

Q3a - What is the purpose of the test whose result is shown in the table below?

Test of Homogeneity of Variances

weight in kg

Levene Statistic	df1	df2	Sig.
.774	2	233	.462

Figure 3: Levene's Test

The Levene's test for homogeneity of variance is to check whether there is a significant difference in variance between the sample groups and this test is used as part of ANOVA.

Q3b - Interpret the results of the table

The value of *Sig.* is 0.462. So, we can accept the null hypothesis and conclude that there is **no significant difference** in the variances between the groups A, B, and C.

ANOVA

weight in kg

	Sum of Squares	df	Mean Square	F	Sig.
Between Groups	166.681	2	83.340	.548	.579
Within Groups	35455.703	233	152.170		
Total	35622.384	235			

Figure 4: ANOVA results

The student decided that the above table showed evidence that there is a difference in weight between groups.

Q4a - Is this statement correct?

Incorrect, since the value of *Sig.* is 0.579, we fail to reject the null hypothesis. There is no significant difference in the weight between the three groups.

Q4b - Interpret in full the results of the table.

Since the null hypothesis is not rejected, we can conclude that there is no significant difference in weight between the three groups.

Multiple Comparisons

Dependent Variable: weight in kg

Scheffe

(I) Group	(J) Group	Mean Difference (I-J)	Std. Error	Sig.	95% Confidence Interval	
					Lower Bound	Upper Bound
A	B	.85191	1.89944	.904	-3.8275	5.5313
	C	2.12766	2.03712	.580	-2.8909	7.1462
B	A	-.85191	1.89944	.904	-5.5313	3.8275
	C	1.27574	2.00010	.816	-3.6516	6.2031
C	A	-2.12766	2.03712	.580	-7.1462	2.8909
	B	-1.27574	2.00010	.816	-6.2031	3.6516

Figure 5: Post-hoc results

Q5 - What is your comment on this next step of the analysis? Was this an appropriate step bearing in mind the ANOVA results?

Since the result of the ANOVA test showed that there is no significant difference in weight between the three groups $p > 0.05$ the post-hoc test is redundant and inappropriate.